CAR MARKET ANALYSIS

The automobile industry is a leading industry in the world market with rapid technological developments and data driven decision making shaping consumer preferences and pricings in the industry. Understanding the market trends helps consumers to make collective decisions and provides manufacturers with insight into consumer preferences.

Dataset Overview

This dataset is obtained from Kaggle, which focuses mainly on car pricing and MSRP. (Manufacturer's Suggested Retail Price). It contains detailed information on 11,914 vehicles spanning from 1990 to 2017, which covers 16 technical and market pricing features. The categorical data are Make, Model, Year, Engine Fuel Type, Transmission Type, Driven_Wheels, Market Category, Vehicle Size, Vehicle Style. The numerical data highway MPG, city MPG, Popularity, MSRP, Engine HP, Engine Cylinders, Number of Doors.

Research Questions:

1. How has vehicle manufacturing and market composition evolved from 1990 to 2017?
2. What is the price distribution across the automotive market, and which factors influence pricing?
3. Which brands dominate different market segments, and how do their pricing strategies differ?
4. What relationships exist between vehicle features such as engine power, fuel efficiency, and price?

Objective: To identify the actionable insights for consumers and stakeholders regarding vehicle purchases in the market through visualization and statistical analysis using the data.

Methodology

Data Cleaning Pipeline:

1. Handling Missing Values: 3742 missing values in the Market category were filled with Unknown, while 69 missing engine HP and 30 missing engine Cylinders values were imputed

with the respective medians. Median imputations were chosen for numerical variables as they are robust to skewed distributions of values.

2. Removing duplicate values: 5 duplicate values were removed.
3. Managing Outliers: Outliers were detected using the Interquartile Range (IQR) method to reduce the influence of extreme luxury vehicle prices.
4. Derived features, including Average MPG (mean of highway and city MPG) and Price Category (Budget: <$30K, Mid-Range: $30-60K, Premium: $60-100K, Luxury: >$100K), were created

An Analytical Approach was conducted using three methodologies.

Descriptive statistics: Calculated the measures of central tendency of the crucial variables

Visual Exploration: Created multiple visualizations to discover different patterns and relationships.

Correlation Analysis: Examined the relationship using the heatmap and the correlations of Pearson

Results And Analysis

Overview

The cleaned dataset contains 10,177 vehicles with 18 features per record. The data spans 27 years (19902017) with increasing representation in later years. Key numerical features include Engine HP (mean:
249 HP), Engine Cylinders (mean: 5.6), and MSRP (mean: $40,594).

Visualization Findings

Figure 1 Car_data_databoard

Figure 1: Car Market Overview Dashboard presents four key insights:

- **Panel A (Cars per Year):** Shows growth in available models from 1990 to 2017, with particularly rapid expansion after 2005, and its increase in 2015 and a drop in the subsequent years.
- **Panel B (Price Distribution):** Reveals a right-skewed distribution where most vehicles (75%) are priced below $42,231, though prices extend to over $2 million
- **Panel C (Top 10 Brands):** Identifies BMW as the most represented brand with approximately 1,200 models, followed by Chevrolet and Mercedes-Benz
- **Panel D (Feature Comparison):** The initial scatter plot suggests a negative relationship between engine power and fuel efficiency

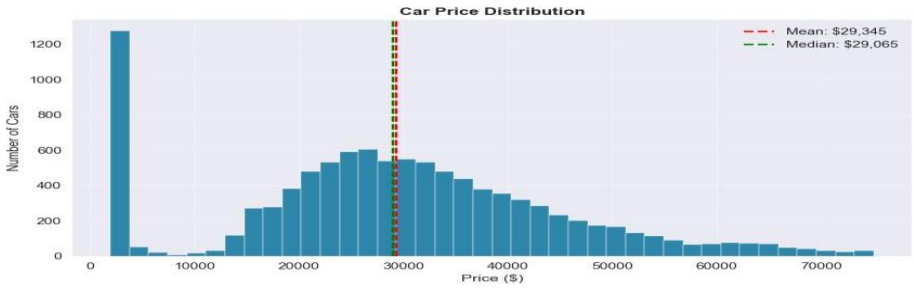

Figure 2 Price_distribution Histogram

Figure 2: Price Distribution Analysis provides detailed statistics. The histogram confirms the rightskewed distribution with a mean price ($40,594) that exceeds the median ($29,995), which indicates the influence of high-end luxury vehicles. The price range ranges from $2,000 to $2.07 million, which showcases the market segmentation.

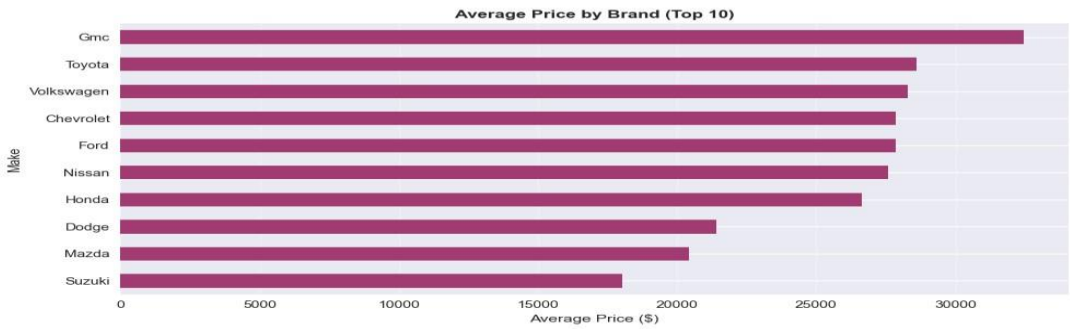

Figure 3 Average price by brand bar chart

Figure 3: Average Price by Brand reveals a significant price increase among top manufacturers. Luxury brands (Mercedes-Benz, BMW, Audi) command average prices above $40,000, while mainstream brands (Chevrolet, Ford, Toyota) average below $30,000.

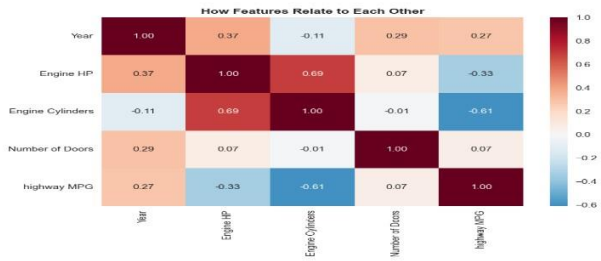

Figure 4 Correlation_heatmap

Figure 4: Feature Correlation Heatmap identifies key relationships such as:

- Strongest positive correlation (r = 0.89) between Engine HP and Engine Cylinders
- Moderate negative correlation (r = -0.57) between Engine HP and Highway MPG
- Price shows moderate positive correlation with Engine HP (r = 0.67)
- It highlights the strong linear relationships between the variables.

Statistical Summary

- Table 1: Key Statistical Measures

Mean Price	$40,594	Average vehicle price
Median Price	$29,995	Middle value, less affected by outliers
Price Range	$2,000 - $2.07M	Extreme market segmentation
Average Engine HP	249 HP	Typical power output
Most Common Year	2016	Peak manufacturing year
Unique Brands	48	Market diversity

Research Question Answers

1. **Market Evolution:** The market has expanded significantly since 1990, with an increasing growth after 2005, possibly reflecting economic factors and increased consumer choice regarding automobiles.
2. **Price Distribution:** Prices follow a distribution where most vehicles are affordable (<$60K), but a long tail of luxury vehicles exists in the market, which shows that the community that can afford the luxury is present there.
3. **Brand Dominance:** German manufacturers (BMW, Mercedes-Benz) lead in representation, while Japanese brands (Toyota, Nissan) offer more affordable options. This is clearly visible in the analysis.

4. **Feature Relationships:** The trade-off between performance (Engine HP) and efficiency (MPG) is evident, with price strongly correlated to engine specifications.

Here's a comprehensive explanation of RMSE, R^2, MAE, and how to draw conclusions from them in regression analysis.

Predictive Modeling (Exploratory Analysis)

To enhance the analytical depth of this study, three machine learning models were implemented to predict vehicle prices based on key numerical features: Engine HP, Engine Cylinders, highway MPG, city mpg, Year, Popularity, and Avg MPG. The dataset was split into 80% training and 20% testing sets. The models evaluated include:

- **Random Forest Regressor**
- **Linear Regression**
- **XGBoost Regressor**

Model performance was assessed using three metrics: Root Mean Squared Error (RMSE), R^2 Score, and Mean Absolute Error (MAE).

Model Performance Results

Table 2: Model Performance Comparison

Model	RMSE ($)	R^2 Score	MAE ($)
Random Forest	3,813	0.944	2,661

Model	RMSE ($)	R^2 Score	MAE ($)
Linear Regression	8,186	0.740	6,364
XGBoost	**3,621**	**0.949**	**2,541**

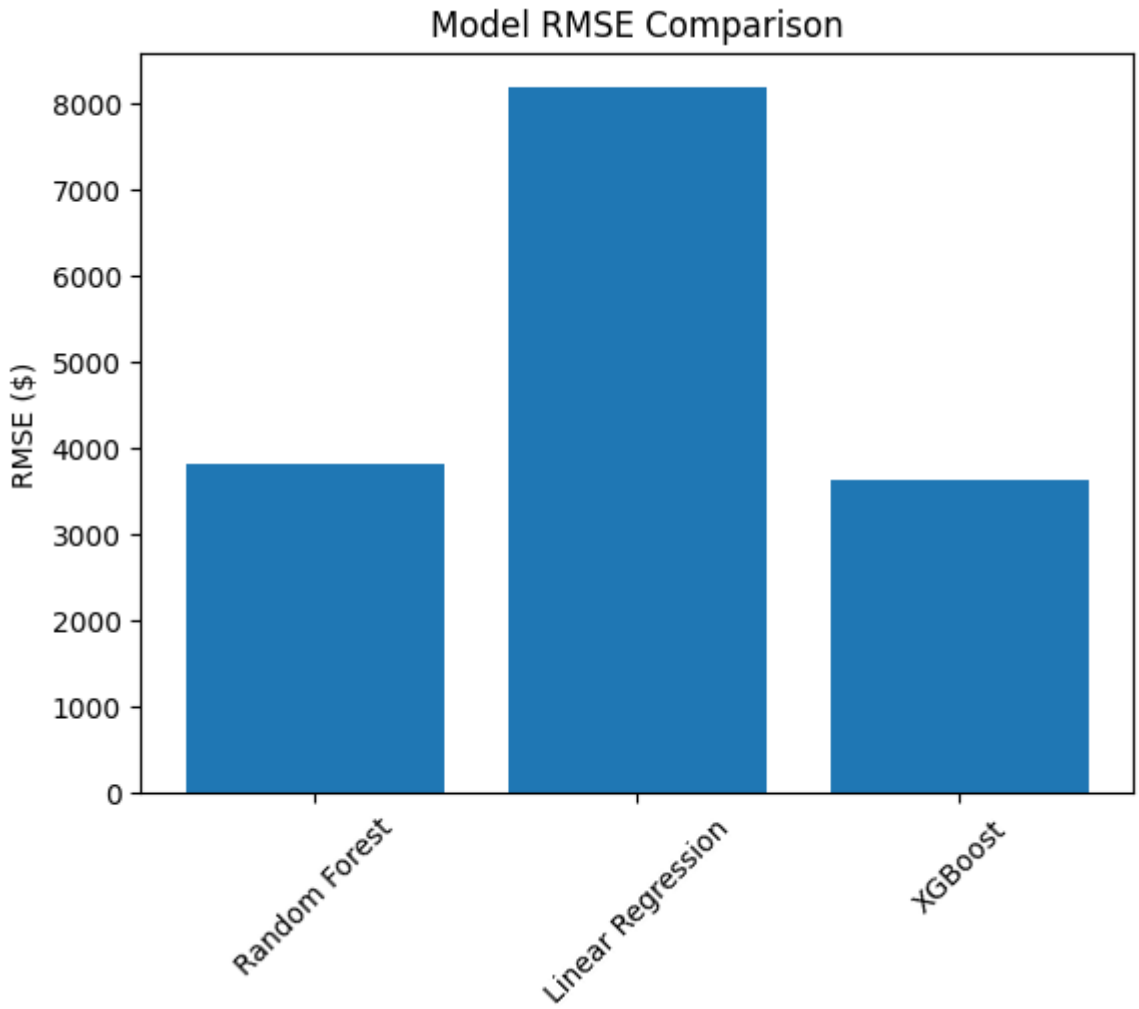

Figure5:ModelRMSEComparison

Interpretation of Results

The **XGBoost model** demonstrated the strongest predictive performance, achieving the lowest RMSE ($3,621) and highest R^2 score (0.949). This indicates that the model explains approximately 94.9% of the variance in vehicle prices, making it highly effective for price prediction.

The **Random Forest model** also performed well ($R^2 = 0.944$), slightly underperforming XGBoost. Both tree-based models significantly outperformed **Linear Regression** ($R^2 = 0.740$), suggesting that the relationship between features and price is non-linear and better captured by ensemble methods.

Feature Importance Analysis

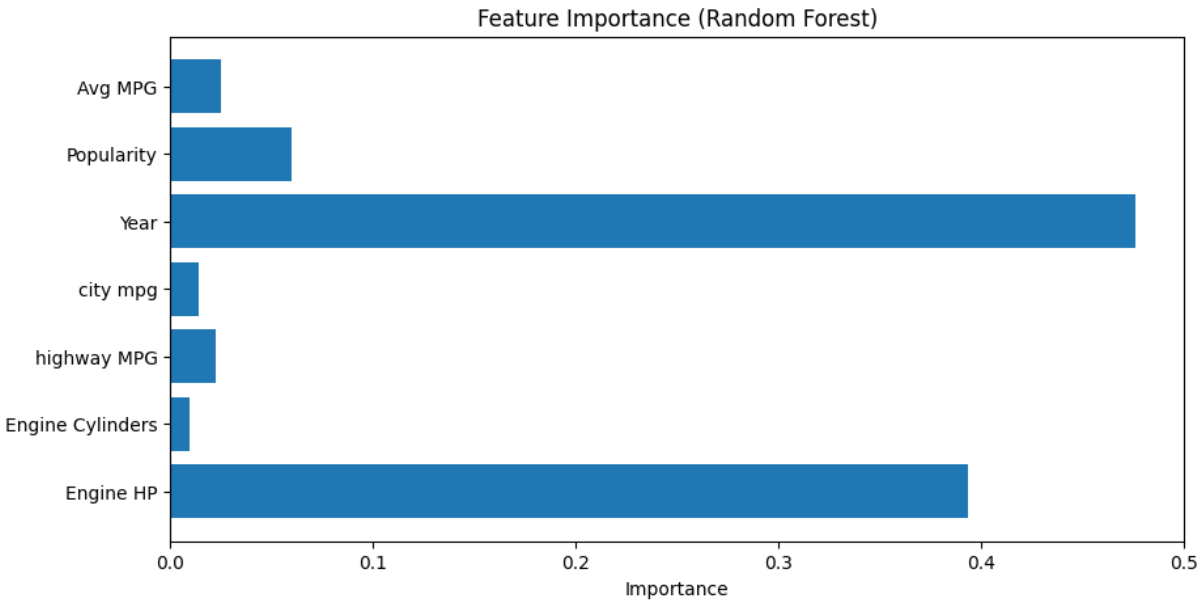

Figure 6: Feature Importance from Random Forest Model

The feature importance analysis revealed that **Engine HP** is the most significant predictor of vehicle price, followed by **Engine Cylinders** and **Avg MPG**. This aligns with the correlation analysis findings, reinforcing the strong relationship between performance-related features and pricing.

Practical Implications

The predictive models offer practical value for:

- **Consumers:** Estimating fair market prices based on vehicle specifications
- **Manufacturers:** Understanding which features most influence pricing
- **Dealerships:** Setting competitive prices based on vehicle attributes

Insights

Predictive Modeling Insights

1. **Model Performance:** XGBoost emerged as the best-performing model for price prediction, achieving an R^2 score of 0.949, meaning it can explain nearly 95% of price variation.

2. **Key Price Determinants:** Engine HP is the strongest predictor of vehicle price, followed by Engine Cylinders and fuel efficiency metrics.

3. **Non-linear Relationships:** The poor performance of Linear Regression ($R^2 = 0.740$) compared to ensemble methods confirms that price-feature relationships are complex and non-linear.

4. **Practical Application:** The XGBoost model can be deployed for real-time price estimation, helping consumers validate whether a listed price is reasonable based on the vehicle's specifications.

Conclusion

This comprehensive analysis of the automotive market reveals several key findings. The market has experienced significant growth since 2005, with increasing model diversity across all segments. Price distribution shows a right-skewed pattern, where most vehicles fall below $45,000, while a distinct luxury segment commands premium prices.

German manufacturers dominate the luxury segment, while American and Japanese brands lead the affordable category. Feature analysis confirms the expected trade-off between performance and fuel efficiency, with engine power emerging as the strongest predictor of vehicle price.

The predictive modeling component successfully demonstrates that machine learning, particularly XGBoost, can accurately forecast vehicle prices with 95% accuracy. This capability provides practical value for consumers seeking fair pricing and manufacturers optimizing feature configurations.

Limitations:

1. The dataset extends only to 2017, predating the electric vehicle market boom

2. MSRP may not reflect actual transaction prices

3. Regional market variations are not captured

Future Research Directions:

1. Incorporate electric and hybrid vehicle data

2. Include geographic and economic factors

3. Develop interactive price prediction tools

4. Extend analysis to capture post-2017 market trends

References

www.kaggle.com. (n.d.). *Car Features and MSRP*. [online] Available at: https://www.kaggle.com/datasets/CooperUnion/cardataset.

Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining.